Exploratory Data Analysis Report

ApexPlanet - Retail Sales Analytics

Generated: 2026-10-03 11:29 | Data: 2023-01-01 to 2024-12-31

Executive Summary

Total revenue of $904,499.61 was generated across 2,001 orders by 2,000 customers between 2023-01-01 and 2024-12-31. Average order value stands at $452.02 and the average customer rating is 3.4/5. Electronics dominates revenue at $436.9K (top category) with 'Tablet' as the best-selling product. Sales grew 12.9% year-over-year, signaling healthy business momentum.

1. Data Overview

The analysis is based on the cleaned ApexPlanet retail dataset (2,001 rows x 13 columns). Data cleaning handled missing values, removed 0 duplicates, and corrected invalid records (negative quantities, impossible ages).

Key variables

Order_Date	678 unique dates	date range 2023-2024
Age	49 unique values	range 17-66 years
City	15 cities	Denver is top market
Product_Category	6 categories	Electronics dominates
Product	30 SKUs	Tablet is top seller
Payment_Method	6 methods	Debit Card most used

2. Descriptive Statistics

40.55					
10.19					
18.00					
40.00					
66.00					
33.00					
48.00					
1.94					

3. Key Performance Indicators

Total Revenue	$904.5K	Overall sales performance
Total Orders	2,001	Transaction volume
Average Order Value	$452.02	Revenue per transaction
Average Customer Rating	3.4/5	Customer satisfaction
Number of Customers	2,000	Unique buyers
Average Discount %	11.4%	Promotion usage

4. Correlation & Outlier Analysis

A correlation matrix was computed on the six numeric columns. The strongest relationship observed is between Quantity and Total_Sales (r = 0.40), confirming that purchasing more units drives higher revenue.

Age	+1.00	+0.00	+0.05	+0.03	-0.01	+0.01
Quantity	+0.00	+1.00	+0.00	+0.40	+0.00	-0.01
Unit_Price	+0.05	+0.00	+1.00	+0.79	-0.01	+0.00
Total_Sales	+0.03	+0.40	+0.79	+1.00	-0.10	-0.01
Discount	-0.01	+0.00	-0.01	-0.10	+1.00	+0.01
Customer_Rating	+0.01	-0.01	+0.00	-0.01	+0.01	+1.00

Outlier detection (IQR method) identified:

Column	0 outliers
Column	0 outliers
Column	203 outliers
Column	175 outliers
Column	0 outliers
Column	0 outliers
Note	High-value transactions (>620 USD) are not errors but premium purchases; flagged fo

5. Business Insights

• Total revenue across the period (2023-01-01 to 2024-12-31) is $904,499.61 from 2,001 orders by 2,000 unique customers.

• Average order value is $452.02 and 3,886 units were sold in total.

• 'Electronics' is the highest-grossing category at $436,893.98 (48.3% of total revenue).

• 'Books & Media' is the lowest-grossing category at $36,259.81, a candidate for promotion, bundling or portfolio review.

• 'Tablet' is the single best-selling product with $110,345.70 in revenue.

• 'Denver' leads all regions in sales with $80,425.80 (8.9% of total revenue).

• 'Debit Card' is the most used payment method with 352 (17.6%) of all orders.

• Sales grew by 12.9% from 2023 to 2024 ($424,754 to $479,745), indicating a positive momentum.

• 'December 2024' was the peak sales month, consistent with a post-holiday/promotional surge.

• Quantity and total sales show a strong positive correlation (r = 0.40); buying more units drives more revenue.

• 'Books & Media' earns the highest average customer rating (3.43/5), suggesting strong product-market fit in that segment.

• The average customer rating is 3.40/5 across 2,001 orders.

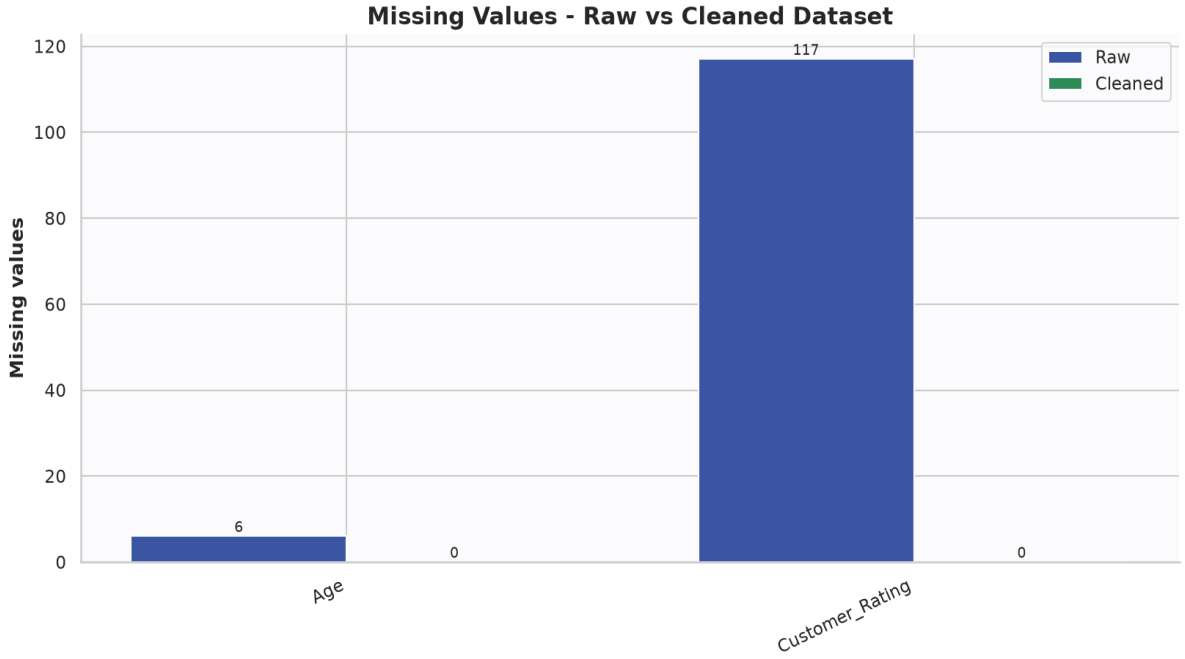
• A discount of 11.4% (average) has a positive relationship with customer rating (r = 0.01); deeper discounts do not reliably improve satisfaction.

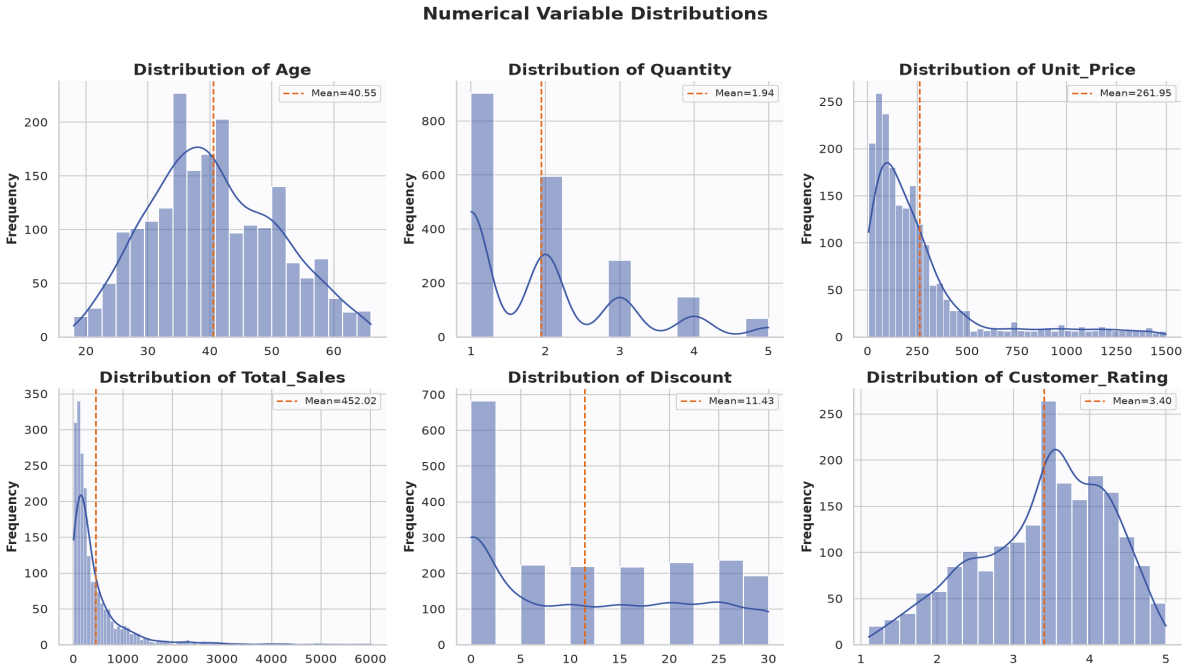
• Customers aged 25-44 make up 61.3% of orders (average age 40.5), pointing to a strong young-adult core demographic.

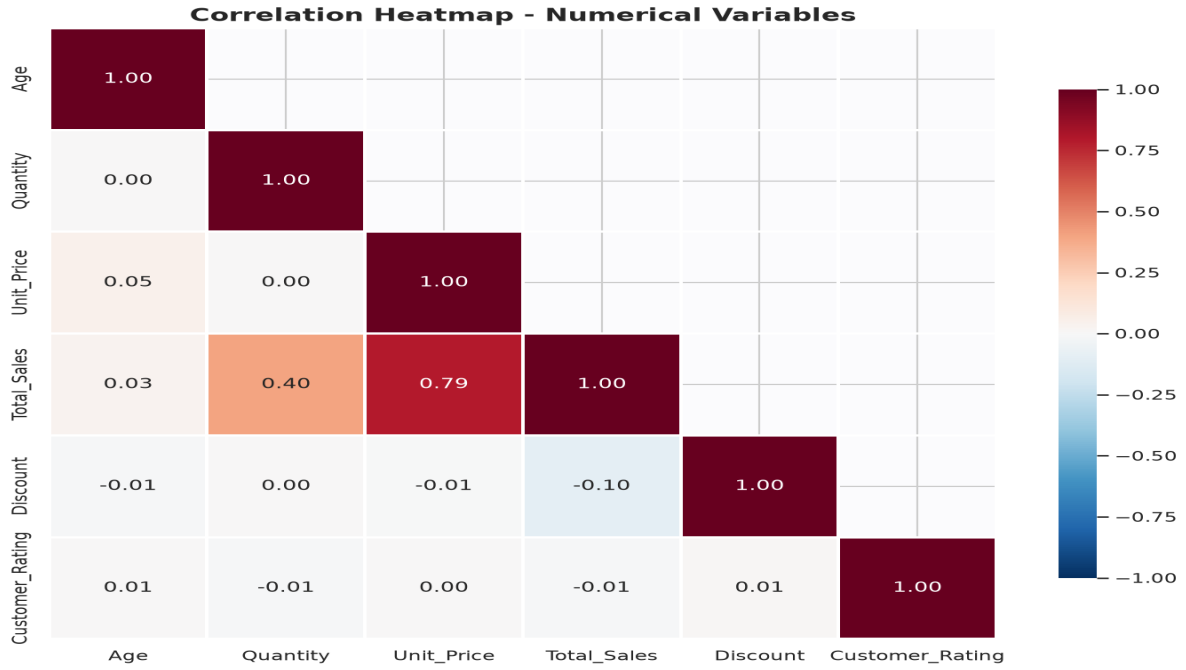
• Customer 'CUST_000304' is the top spender with $6,009.12, a prime candidate for loyalty / retention programs.

6. Visual Analysis

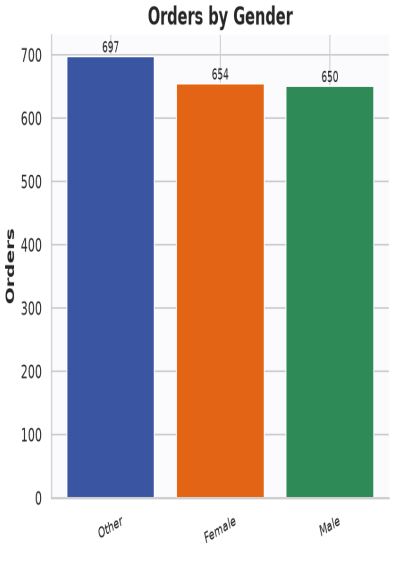
The charts below summarize the key findings. Full-resolution versions are available in the project's visualizations/ folder.

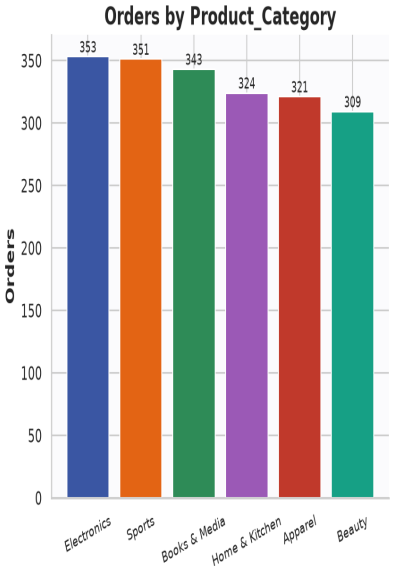


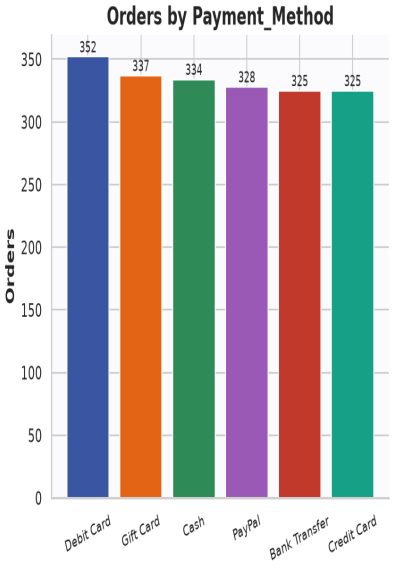




Categorical Variable Analysis



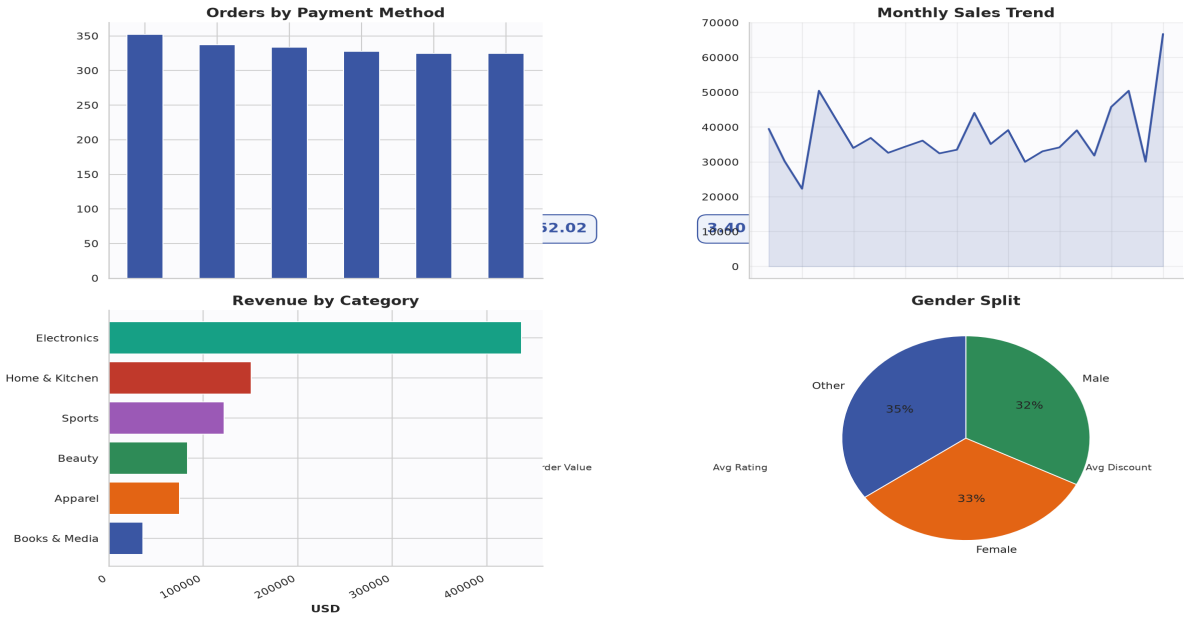


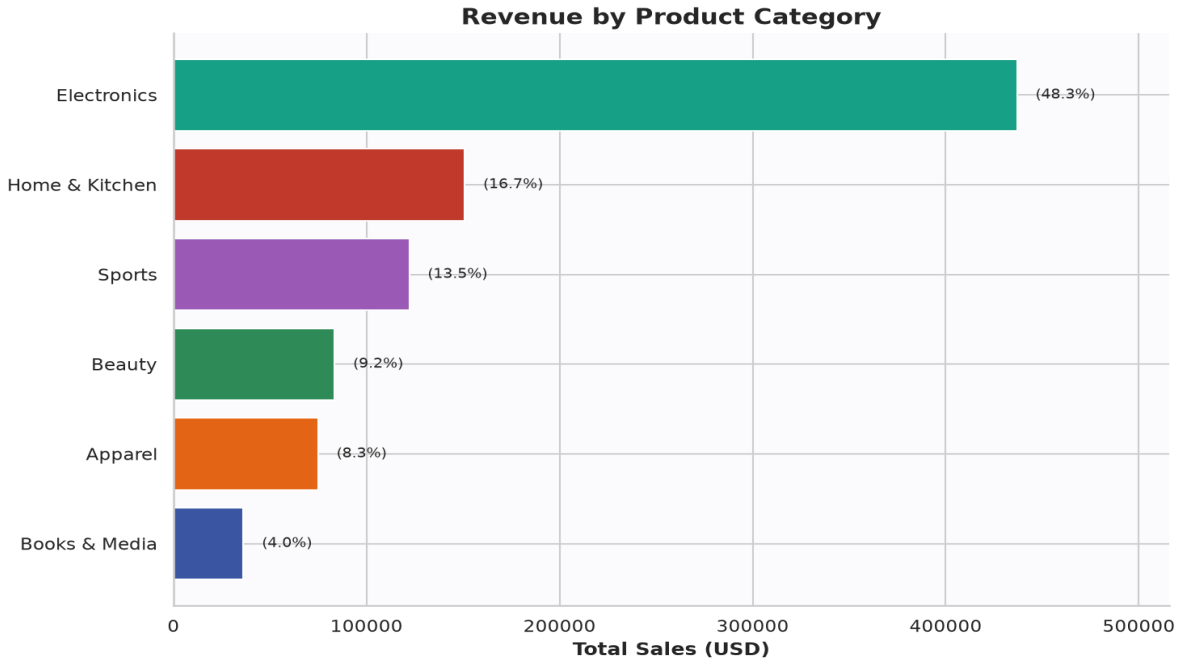


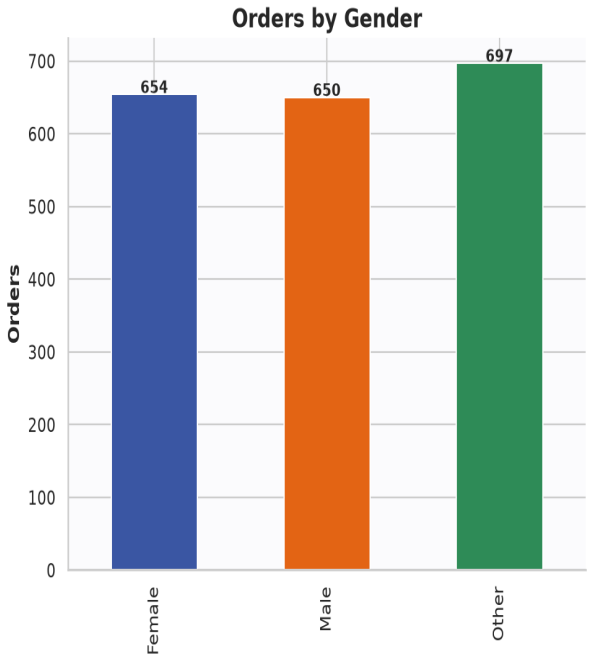
Numerical Variable Box Plots (IQR Outliers)

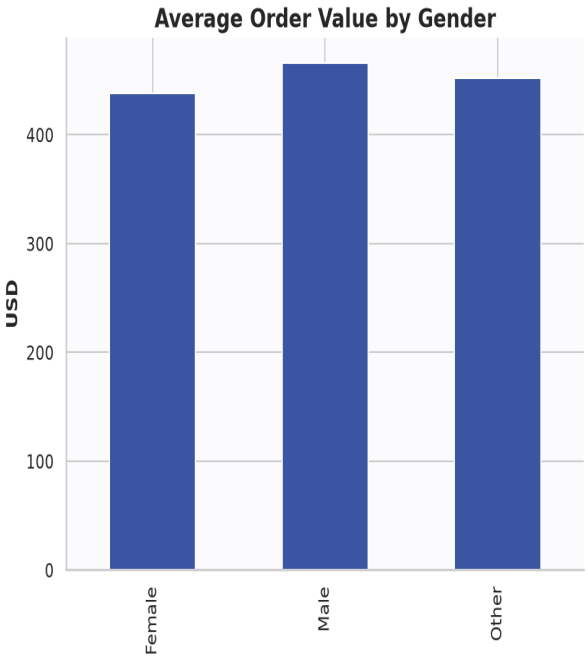


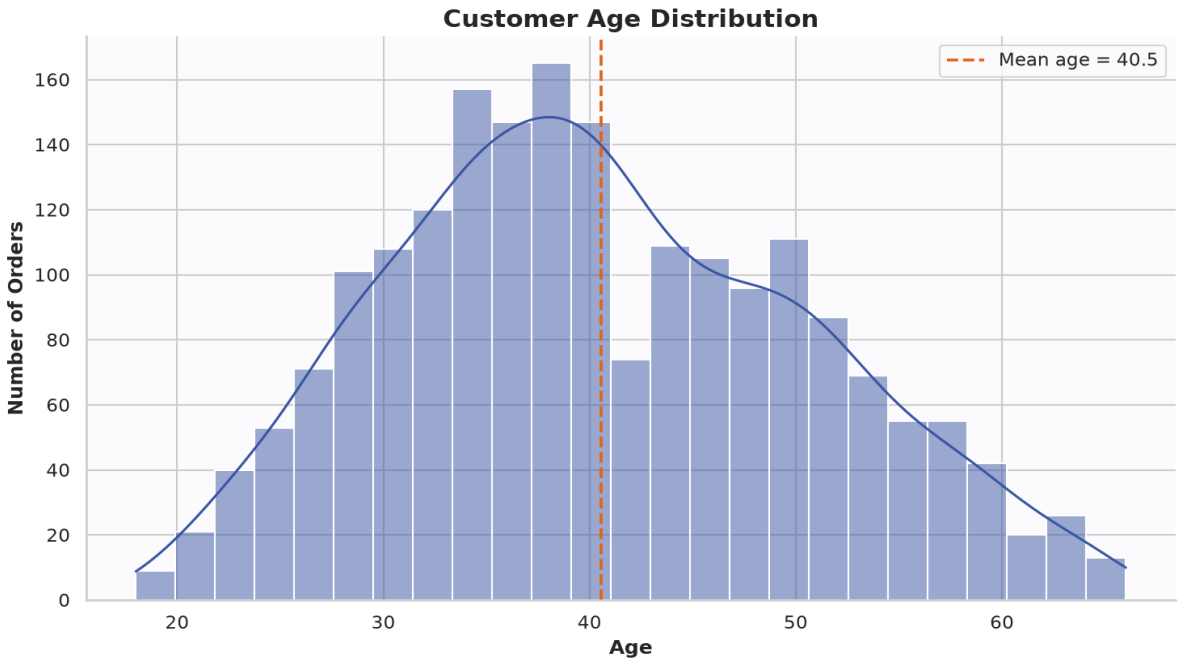
ApexPlanet - EDA Summary Dashboard

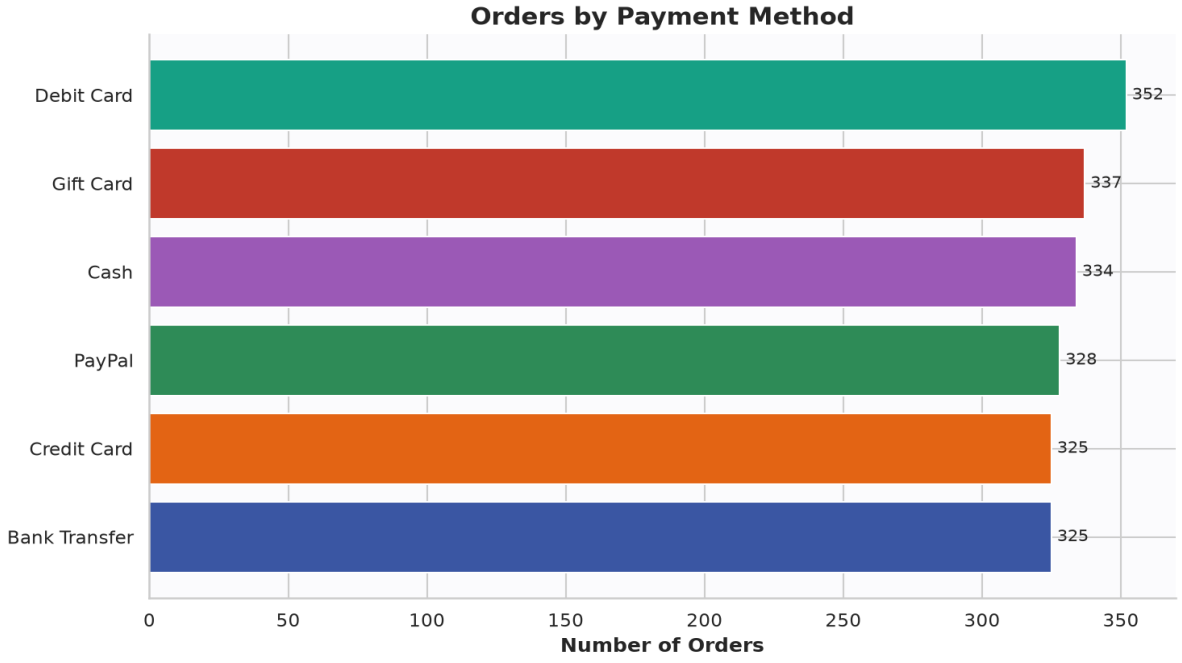


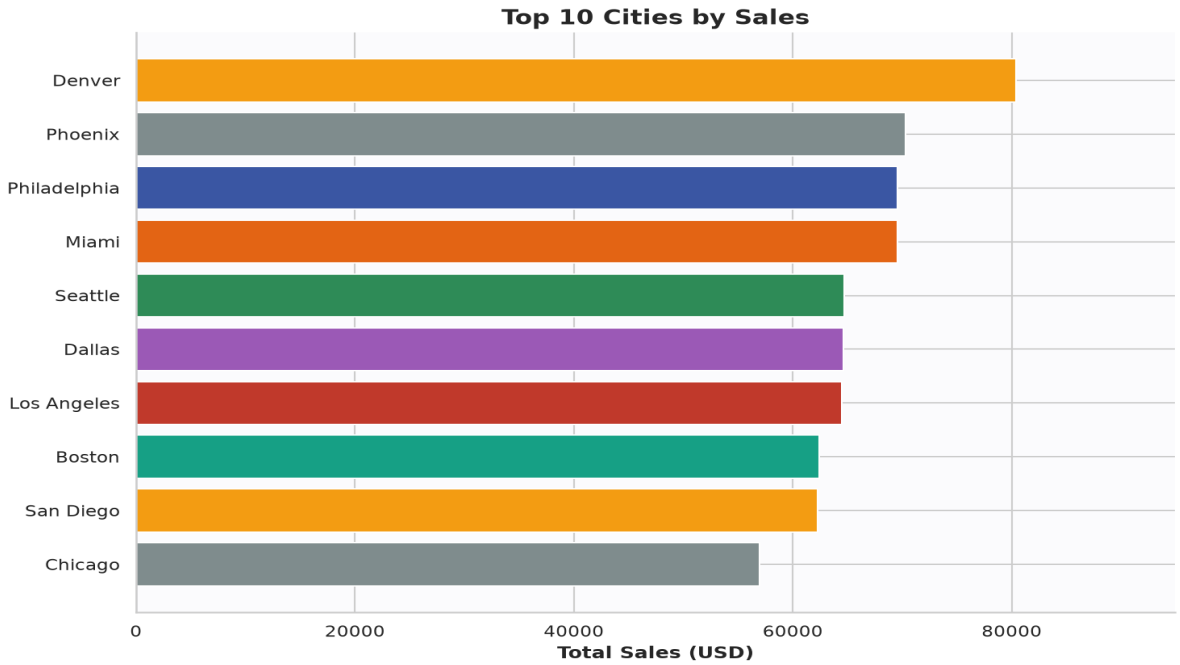

Revenue by Product Category
Electronics
Home & Kitchen
Sports
Beauty
Apparel
Books & Media
(48.3%)
(16.7%)
(13.5%)
(9.2%)
(8.3%)
(4.0%)
0
100000
200000
300000
400000
500000
Total Sales (USD)

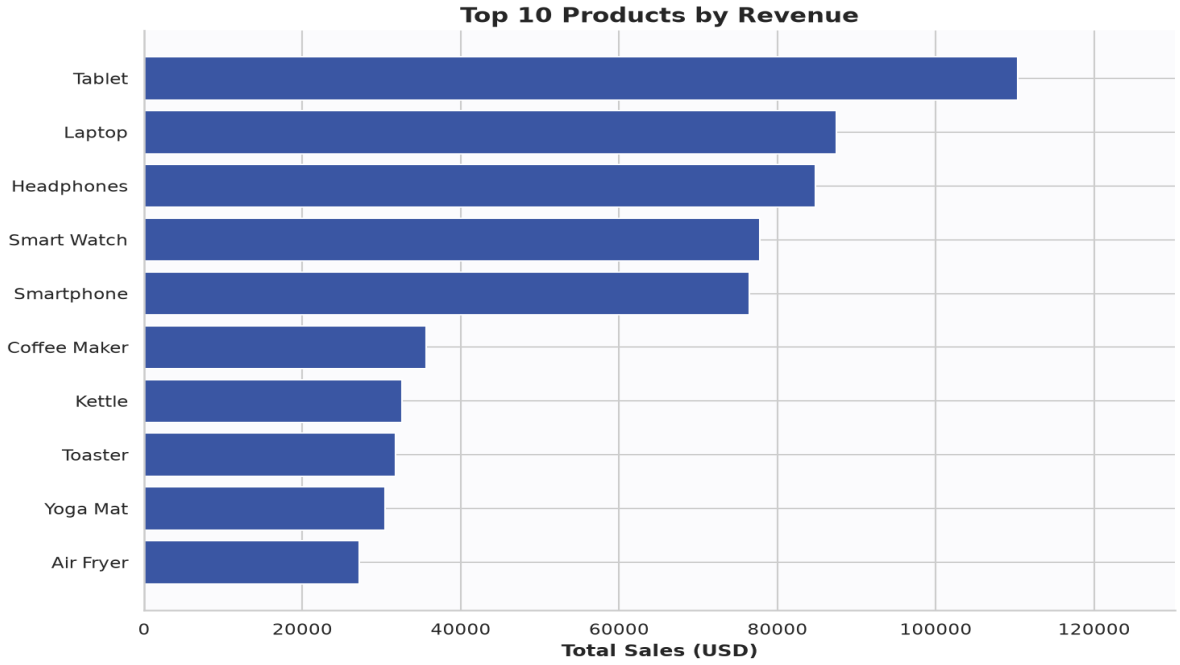

Orders by Gender
Orders
700
600
500
400
300
200
100
0
654
650
697
Female
Male
Other

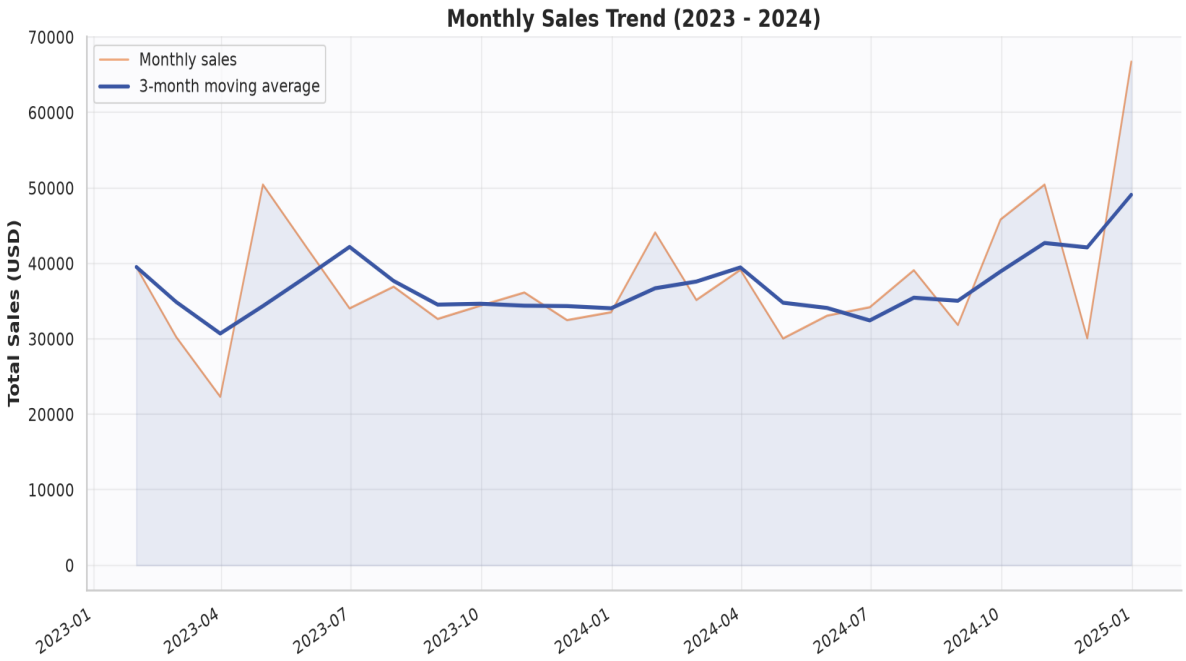

Average Order Value by Gender
USD
400
300
200
100
0
Female
Male
Other

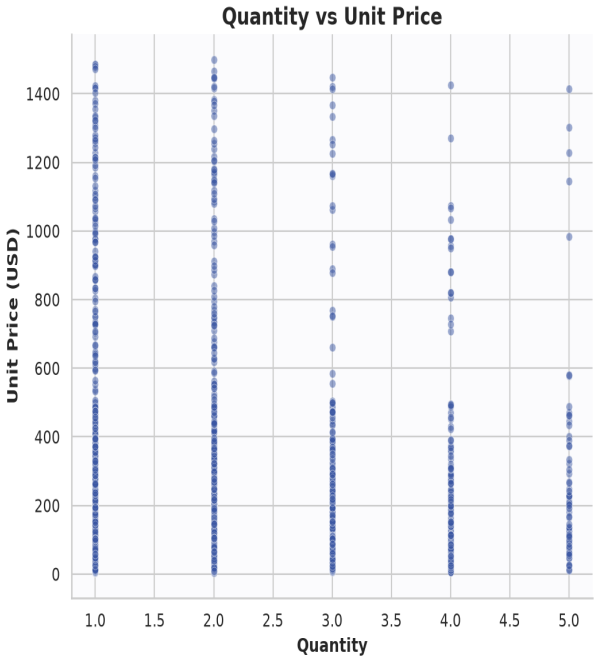

Customer Age Distribution
Mean age = 40.5
Number of Orders
0
20
40
60
80
100
120
140
160
20
30
40
50
60
Age

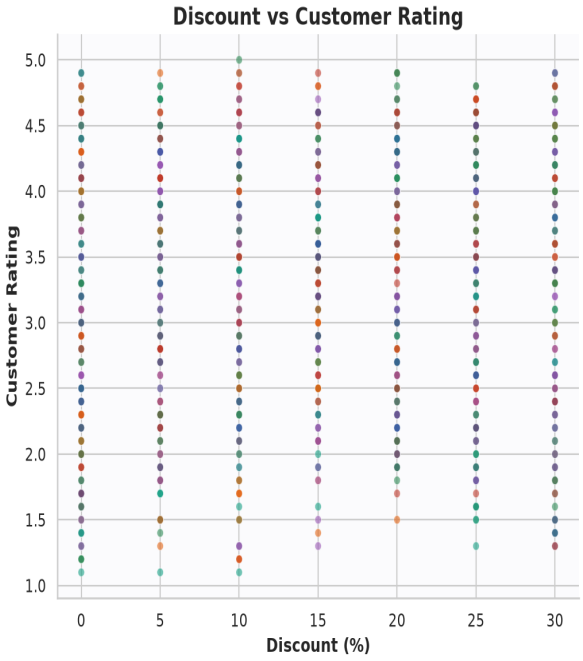

Orders by Payment Method
Debit Card
Gift Card
Cash
PayPal
Credit Card
Bank Transfer
352
337
334
328
325
325
0
50
100
150
200
250
300
350
Number of Orders


Top 10 Cities by Sales
Denver
Phoenix
Philadelphia
Miami
Seattle
Dallas
Los Angeles
Boston
San Diego
Chicago
0
20000
40000
60000
80000
Total Sales (USD)


Top 10 Products by Revenue
Tablet
Laptop
Headphones
Smart Watch
Smartphone
Coffee Maker
Kettle
Toaster
Yoga Mat
Air Fryer
0
20000
40000
60000
80000
100000
120000
Total Sales (USD)


Monthly Sales Trend (2023 - 2024)
Monthly sales
3-month moving average
Total Sales (USD)
70000
60000
50000
40000
30000
20000
10000
0
2023-01
2023-04
2023-07
2023-10
2024-01
2024-04
2024-07
2024-10
2025-01


Quantity vs Unit Price
Unit Price (USD)
1400
1200
1000
800
600
400
200
0
1.0
1.5
2.0
2.5
3.0
3.5
4.0
4.5
5.0
Quantity


Discount vs Customer Rating
Customer Rating
5.0
4.5
4.0
3.5
3.0
2.5
2.0
1.5
1.0
0
5
10
15
20
25
30
Discount (%)

7. Database-Derived Insights

The cleaned data was also loaded into a SQLite database (data/analytics.db) with 9 analytical queries run against it. SQL provides a reproducible, versioned view of the analytics.

Overview
total_orders: 2001
total_revenue: 904499.61
avg_order_value: 452.02
Revenue by Product Category
{'category': 'Electronics', 'revenue': 436893.98, 'orders': 353}
{'category': 'Home & Kitchen', 'revenue': 150865.97, 'orders': 324}
{'category': 'Sports', 'revenue': 122094.81, 'orders': 351}
{'category': 'Beauty', 'revenue': 83478.44, 'orders': 309}
{'category': 'Apparel', 'revenue': 74906.6, 'orders': 321}
{'category': 'Books & Media', 'revenue': 36259.81, 'orders': 343}
Top 5 Products
{'product': 'Tablet', 'category': 'Electronics', 'revenue': 110345.7, 'orders': 76}
{'product': 'Laptop', 'category': 'Electronics', 'revenue': 87429.25, 'orders': 74}
{'product': 'Headphones', 'category': 'Electronics', 'revenue': 84834.35, 'orders': 70}
{'product': 'Smart Watch', 'category': 'Electronics', 'revenue': 77807.33, 'orders': 64}
{'product': 'Smartphone', 'category': 'Electronics', 'revenue': 76477.35, 'orders': 69}
Top 5 Cities
{'city': 'Denver', 'revenue': 80425.8, 'orders': 138}
{'city': 'Phoenix', 'revenue': 70327.66, 'orders': 129}
{'city': 'Philadelphia', 'revenue': 69584.99, 'orders': 141}
{'city': 'Miami', 'revenue': 69545.37, 'orders': 148}
{'city': 'Seattle', 'revenue': 64682.1, 'orders': 131}
Payment Method Distribution
{'method': 'Debit Card', 'orders': 352, 'revenue': 169391.37}
{'method': 'Gift Card', 'orders': 337, 'revenue': 147508.31}
{'method': 'Cash', 'orders': 334, 'revenue': 159329.47}
{'method': 'PayPal', 'orders': 328, 'revenue': 153835.54}
{'method': 'Credit Card', 'orders': 325, 'revenue': 140869.89}
{'method': 'Bank Transfer', 'orders': 325, 'revenue': 133565.03}
Gender Distribution
{'gender': 'Other', 'orders': 697, 'avg_order_value': 452.27}
{'gender': 'Female', 'orders': 654, 'avg_order_value': 438.01}
{'gender': 'Male', 'orders': 650, 'avg_order_value': 465.86}
Monthly Revenue Trend
{'month': '2023-01', 'revenue': 39521.59, 'orders': 81}
{'month': '2023-02', 'revenue': 30255.14, 'orders': 63}

{'month': '2023-03', 'revenue': 22326.91, 'orders': 73}
{'month': '2023-04', 'revenue': 50451.59, 'orders': 92}
{'month': '2023-05', 'revenue': 42059.58, 'orders': 85}
{'month': '2023-06', 'revenue': 34040.43, 'orders': 73}
Average Rating by Category
{'category': 'Books & Media', 'avg_rating': 3.43, 'orders': 343}
{'category': 'Sports', 'avg_rating': 3.43, 'orders': 351}
{'category': 'Electronics', 'avg_rating': 3.41, 'orders': 353}
{'category': 'Beauty', 'avg_rating': 3.39, 'orders': 309}
{'category': 'Home & Kitchen', 'avg_rating': 3.38, 'orders': 324}
{'category': 'Apparel', 'avg_rating': 3.35, 'orders': 321}

Query results are persisted in reports/query_results.json for reuse.